



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 8: Consumer Theory VI

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ICS2523 Microeconomics

Lecture 8

Last lecture we

- Reviewed the **Slutsky equation** and its derivation
- Did an exercise to practise these new concepts

In today's lecture, we will

- Review the homework from last lecture
- Learn about the **individual demand**

Homework L7: Cobb-Douglas

If the utility function of an individual is $U(x, y) = x^{0.4}y^{0.6}$ find his/hers:

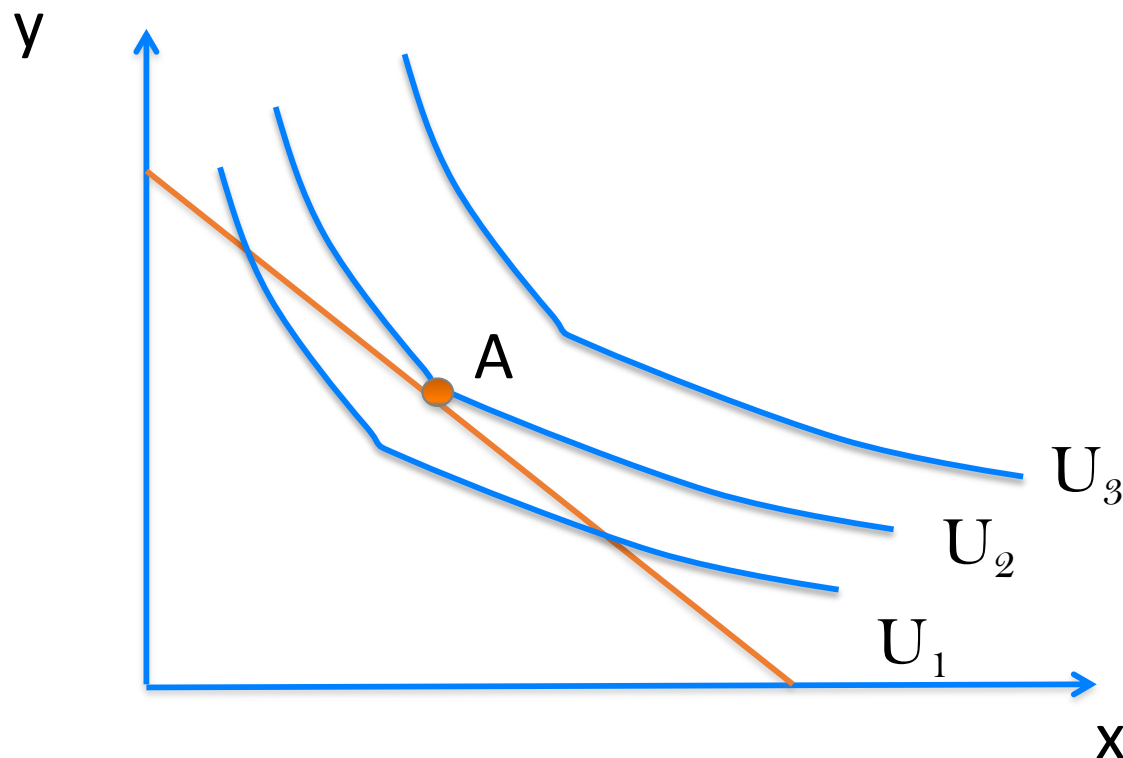
- Find his Marshallian demand function. Verify that it is homogenous of degree zero in prices and income.
- Find his Indirect utility function. Verify that it is homogenous of degree zero in prices and income.
- Find his Hicksian demand function. Verify that it is homogenous of degree zero in prices.
- Find his Expenditure function.

Finally,

- Verify that Roy's identity and Shephard's lemma.
- Verify Slutsky's equation.

Homework L7: solution

Step 1: Before attempting to solve the problem mathematically, **check (graphically) if it is an interior solution or corner solution**



A Cobb-Douglas utility function given by
 $U(x, y) = x^\alpha y^\beta$,
where $\alpha + \beta = 1$
 $0.4 + 0.6 = 1$

$$\rightarrow MRS = \frac{P_x}{P_y}$$

Homework L7: solution

Step 2: Now that we know that we can look for the optimum consumption bundle mathematically, we use the utility maximization problem:

$$\text{Max}_{x,y} U(x, y) = x^\alpha y^\beta, \text{ where } \alpha + \beta = 1$$

Subject to:

$$P_x x + P_y y = I$$

Thus, the Lagrangian is

$$\mathcal{L}(x, y, \lambda) = x^\alpha y^\beta + \lambda(I - P_x x - P_y y)$$

Homework L7: solution

Step 3: Derive the first-order conditions

$$\mathcal{L}(x, y, \lambda) = x^\alpha y^\beta + \lambda(I - P_x x - P_y y)$$

$$\frac{\partial \mathcal{L}}{\partial x} = \alpha x^{\alpha-1} y^\beta - \lambda P_x = 0 \rightarrow \lambda = \frac{\alpha x^{\alpha-1} y^\beta}{P_x} \text{ (eq 1)}$$

$$\frac{\partial \mathcal{L}}{\partial y} = \beta x^\alpha y^{\beta-1} - \lambda P_y = 0 \rightarrow \lambda = \frac{\beta x^\alpha y^{\beta-1}}{P_y} \text{ (eq 2)}$$

Since it is the same $\lambda \rightarrow \frac{\alpha x^{\alpha-1} y^\beta}{P_x} = \frac{\beta x^\alpha y^{\beta-1}}{P_y} \rightarrow \frac{\alpha y}{\beta x} = \frac{P_x}{P_y} (*)$

Homework L7: solution

$$\mathcal{L}(x, y, \lambda) = x^\alpha y^\beta + \lambda(I - P_x x - P_y y)$$

The last first-order condition is:

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - P_x x - P_y y = 0$$

Replacing (*) in this condition yields

$$I - P_x x - P_y \frac{P_x \beta x}{\alpha P_y} = 0$$

$$P_x x \left(1 + \frac{\beta}{\alpha}\right) = I$$

$$x_m^* = \frac{\alpha}{\alpha + \beta} \frac{I}{P_x}$$

$$x_m^* = 0.4 \frac{I}{P_x}, y_m^* = 0.6 \frac{I}{P_y}$$

Homework L7: solution

Another route for Step 3: remember the optimality condition

$$MRS = \frac{P_x}{P_y}$$

$$\frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$

$$\frac{\alpha x^{\alpha-1} y^{\beta}}{\beta x^{\alpha} y^{\beta-1}} = \frac{P_x}{P_y}$$

$$\frac{\alpha y}{\beta x} = \frac{P_x}{P_y} (*)$$

Homework L7: solution

Replacing (*) in the Budget constraint yields:

$$I - P_x x - P_y y = 0$$

$$I - P_x x - P_y \frac{P_x \beta x}{\alpha P_y} = 0$$

$$P_x x \left(1 + \frac{\beta}{\alpha} \right) = I$$

$$x_m^* = \frac{\alpha}{\alpha + \beta} \frac{I}{P_x}$$

$$x_m^* = 0.4 \frac{I}{P_x}, y_m^* = 0.6 \frac{I}{P_y}$$

Homework L7: solution

Are these Marshallian demands homogenous of degree zero in prices and income?

$$x_m^*(aP_x, aP_y, aI) = a^0 x_m^*(P_x, P_y, I)$$

$$x_m^*(aP_x, aP_y, aI) = 0.4 \frac{(aI)}{aP_x} = x_m^*(P_x, P_y, I)$$

Homework L7: solution

The indirect utility function will be:

$$V(P_x, P_y, I) = x^*{}^\alpha y^*{}^\beta = (0.4 \frac{I}{P_x})^{0.4} (0.6 \frac{I}{P_y})^{0.6}$$

$$V(P_x, P_y, I) = (\frac{0.4}{P_x})^{0.4} (\frac{0.6}{P_y})^{0.6} I$$

Is this indirect utility function homogenous of degree zero in prices and income?

$$V(aP_x, aP_y, aI) = a^0 V(P_x, P_y, I)$$

$$V(aP_x, aP_y, aI) = (aI) (\frac{0.4}{aP_x})^{0.4} (\frac{0.6}{aP_y})^{0.6} = a^0 I (\frac{0.4}{P_x})^{0.4} (\frac{0.6}{P_y})^{0.6}$$

Homework L7: solution

Dual problem:

Now, we are minimizing expenditure subject to a certain level of Utility.

We already know that it is an interior solution, so we will go through the solution mathematically.

$$\text{Min}_{x,y} P_x x + P_y y$$

Subject to:

$$x^\alpha y^\beta = \bar{U}(\mu)$$

$$\mathcal{L}(x, y, \mu) = P_x x + P_y y + \mu(\bar{U} - x^\alpha y^\beta)$$

Homework L7: solution

$$\mathcal{L}(x, y, \mu) = P_x x + P_y y + \mu(\bar{U} - x^\alpha y^\beta)$$

First order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = P_x - \mu \alpha x^{\alpha-1} y^\beta = 0 \rightarrow \mu = \frac{P_x}{\alpha x^{\alpha-1} y^\beta} \text{ (eq 1)}$$

$$\frac{\partial \mathcal{L}}{\partial y} = P_y - \mu \beta x^\alpha y^{\beta-1} = 0 \rightarrow \mu = \frac{P_y}{\beta x^\alpha y^{\beta-1}} \text{ (eq 2)}$$

$$\frac{\partial \mathcal{L}}{\partial \mu} = \bar{U} - x^\alpha y^\beta = 0 \text{ (eq 3)}$$

Homework L7: solution

From eq.1 and eq. 2, we have

$$\frac{P_y}{\beta x^\alpha y^{\beta-1}} = \frac{P_x}{\alpha x^{\alpha-1} y^\beta}$$
$$x = y \frac{P_y \alpha}{P_x \beta}$$

In eq. 3 yields

$$\left(y \frac{P_y \alpha}{P_x \beta} \right)^\alpha y^\beta = \bar{U}$$

$$y_h^* = \bar{U} \left(\frac{P_x \beta}{P_y \alpha} \right)^\alpha, x_h^* = \bar{U} \left(\frac{P_y \alpha}{P_x \beta} \right)^\beta$$

Homework L7: solution

$$y_h = \bar{U} \left(\frac{P_x \beta}{P_y \alpha} \right)^\alpha, x_h = \bar{U} \left(\frac{P_y \alpha}{P_x \beta} \right)^\beta$$

To test if these hicksian demands are homogenous of degree zero in Prices:

$$y_h = \bar{U} \left(\frac{tP_x \beta}{tP_y \alpha} \right)^\alpha = \bar{U} t^0 \left(\frac{P_x \beta}{P_y \alpha} \right)^\alpha$$

$$x_h = \bar{U} \left(\frac{tP_y \alpha}{tP_x \beta} \right)^\beta = \bar{U} t^0 \left(\frac{P_y \alpha}{P_x \beta} \right)^\beta$$

Homework L7: solution

The expenditure function is:

$$E(P_x, P_y, U) = P_x x_h^* + P_y y_h^* = P_x \cdot \bar{U} \left(\frac{P_y \alpha}{P_x \beta} \right)^\beta + P_y \cdot \bar{U} \left(\frac{P_x \beta}{P_y \alpha} \right)^\alpha$$

$$E(P_x, P_y, U) = \bar{U} (P_x)^{0.4} (P_y)^{0.6} \left[\left(\frac{\alpha}{\beta} \right)^\beta + \left(\frac{\beta}{\alpha} \right)^\alpha \right]$$

$$E(P_x, P_y, U) = \bar{U} (P_x)^{0.4} (P_y)^{0.6} \left[\frac{\alpha^\beta \alpha^\alpha + \beta^\beta \beta^\alpha}{\alpha^\alpha \beta^\beta} \right]$$

$$E(P_x, P_y, U) = \bar{U} (P_x)^{0.4} (P_y)^{0.6} \left[\frac{\alpha + \beta}{\alpha^\alpha \beta^\beta} \right] = \bar{U} (P_x)^{0.4} (P_y)^{0.6} \left[\frac{1}{\alpha^\alpha \beta^\beta} \right]$$

Homework L7: solution

$$\text{Roy's identity: } x_m^* = - \frac{\frac{\partial V(P_x, P_y, I)}{\partial P_x}}{\frac{\partial V(P_x, P_y, I)}{\partial I}}$$

$$\frac{\partial V(P_x, P_y, I)}{\partial P_x} = \frac{\partial \left(\left(\frac{0.4}{P_x} \right)^{0.4} \left(\frac{0.6}{P_y} \right)^{0.6} I \right)}{\partial P_x} = -0.4^{0.4} 0.4 (P_x)^{-1.4} \left(\frac{0.6}{P_y} \right)^{0.6} I$$

$$\frac{\partial V(P_x, P_y, I)}{\partial I} = \frac{\partial \left(\left(\frac{0.4}{P_x} \right)^{0.4} \left(\frac{0.6}{P_y} \right)^{0.6} I \right)}{\partial I} = \left(\frac{0.4}{P_x} \right)^{0.4} \left(\frac{0.6}{P_y} \right)^{0.6}$$

Thus,

$$x_m^* = - \frac{\frac{\partial V(P_x, P_y, I)}{\partial P_x}}{\frac{\partial V(P_x, P_y, I)}{\partial I}} = - \frac{-0.4^{1.4} (P_x)^{-1.4} \left(\frac{0.6}{P_y} \right)^{0.6} I}{\left(\frac{0.4}{P_x} \right)^{0.4} \left(\frac{0.6}{P_y} \right)^{0.6}} = \frac{0.4I}{P_x}$$

Example L7: solution

Shephard's Lemma: $x_h^* = \frac{\partial E(P_x, P_y, U)}{\partial P_x}$

$$E(P_x, P_y, U) = \bar{U}(P_x)^{0.4} (P_y)^{0.6} \left[\frac{1}{\alpha^\alpha \beta^\beta} \right]$$

$$\frac{\partial E(P_x, P_y, U)}{\partial P_x} = 0.4 \bar{U}(P_x)^{-0.6} (P_y)^{0.6} \left[\frac{1}{0.4^{0.4} 0.6^{0.6}} \right]$$

$$= x_h^*$$

$$\frac{\partial E(P_x, P_y, U)}{\partial P_x} = \bar{U} \left(\frac{0.4 P_y}{0.6 P_x} \right)^{0.6} = x_h^*$$

Example L7: solution

Slutsky Equation:

$$\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial P_j} = \underbrace{\frac{\partial h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))}{\partial P_j}}_{\text{SUBSTITUTION EFFECT}} - \underbrace{\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial I} x_j(P_1, \dots, P_n, I)}_{\text{INCOME EFFECT}}$$

$$\frac{\partial x_h^*}{\partial P_x} = \frac{\partial \left(\bar{U} \left(\frac{0.4P_y}{0.6P_x} \right)^{0.6} \right)}{\partial P_x} = \bar{U} \left(\frac{0.4P_y}{0.6} \right)^{0.6} (-0.6(P_x)^{-1.6})$$

$$\frac{\partial x_h^*}{\partial P_x} = -\bar{U} (0.4P_y)^{0.6} 0.6^{0.4} (P_x)^{-1.6}$$

Example L7: solution

$$\frac{\partial x_m^*}{\partial I} = \frac{\partial(0.4 \frac{I}{P_x})}{\partial I} = \frac{0.4}{P_x}$$

$$\frac{\partial x_m^*}{\partial P_x} = \frac{\partial(0.4 \frac{I}{P_x})}{\partial P_x} = -I \frac{0.4}{(P_x)^2}$$

Example L7: solution

$$\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial P_j} = \frac{\partial h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))}{\partial P_j} - \frac{\partial x_i(P_1, \dots, P_n, I)}{\partial I} x_j(P_1, \dots, P_n, I)$$

$$-I \frac{0.4}{(P_x)^2} = -\bar{U} (0.4 P_y)^{0.6} 0.6^{0.4} (P_x)^{-1.6} - \frac{0.4}{P_x} 0.4 \frac{I}{P_x}$$

Replacing the indirect utility function in $\bar{U}$:

$$-I \frac{0.4}{(P_x)^2} = -\left(\frac{0.4}{P_x}\right)^{0.4} \left(\frac{0.6}{P_y}\right)^{0.6} I (0.4 P_y)^{0.6} 0.6^{0.4} (P_x)^{-1.6} - \frac{0.4}{P_x} 0.4 \frac{I}{P_x}$$

$$-I \frac{0.4}{(P_x)^2} = -(0.4)^{0.4} (0.6)^{0.6} I (0.4)^{0.6} 0.6^{0.4} (P_x)^{-2} - I \frac{0.4^2}{(P_x)^2}$$

$$-I \frac{0.4}{(P_x)^2} = -0.4 \cdot 0.6 I (P_x)^{-2} - I \frac{0.4^2}{(P_x)^2} = -\frac{0.4I}{(P_x)^2} \{0.4 + 0.6\} = -I \frac{0.4}{(P_x)^2}$$

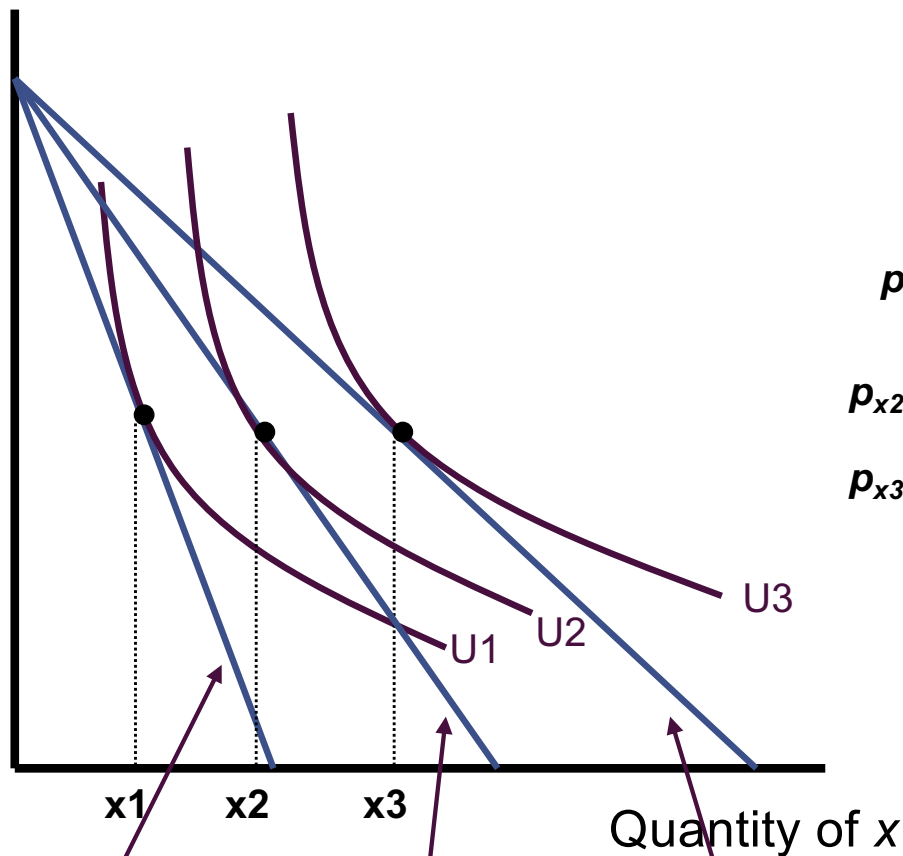
Individual Demand

Consumer Theory

10. Individual Demand

Marshallian demand: Let's see what happens when we vary P_x (ceteris paribus)

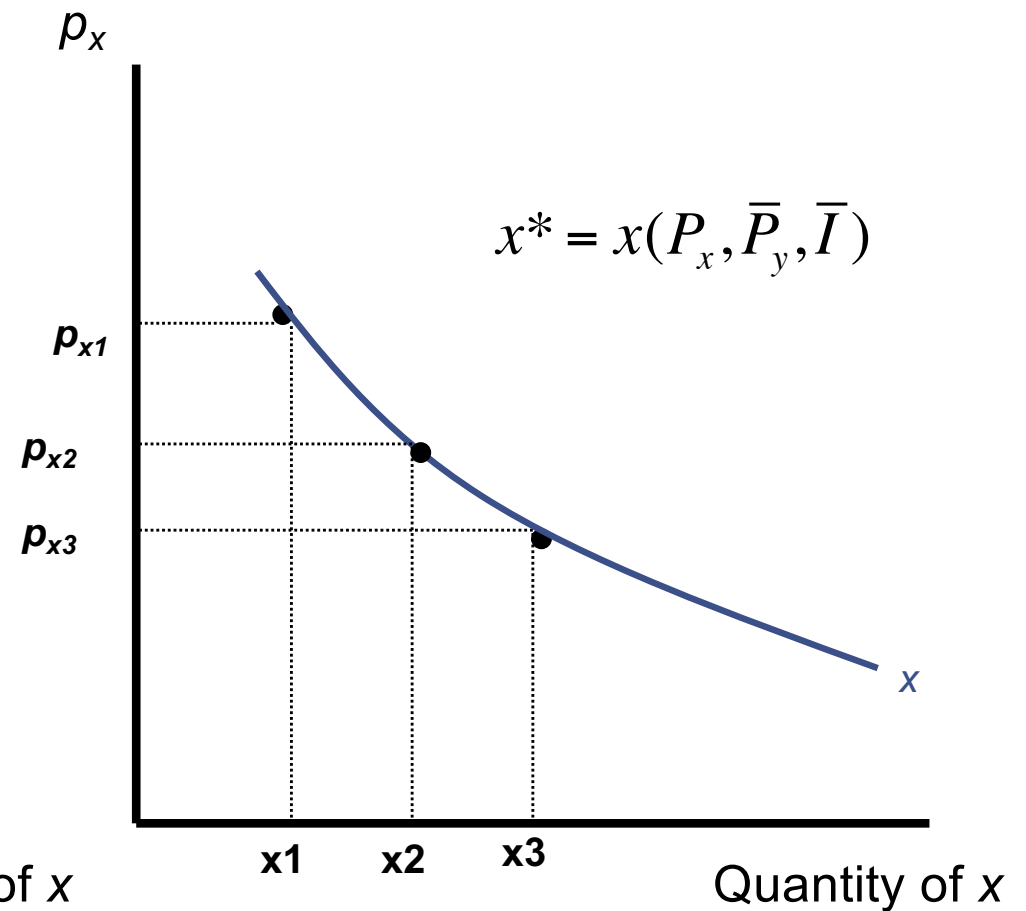
Quantity of y



$$I = p_{x1}x + p_y y$$

$$I = p_{x2}x + p_y y$$

$$I = p_{x3}x + p_y y$$

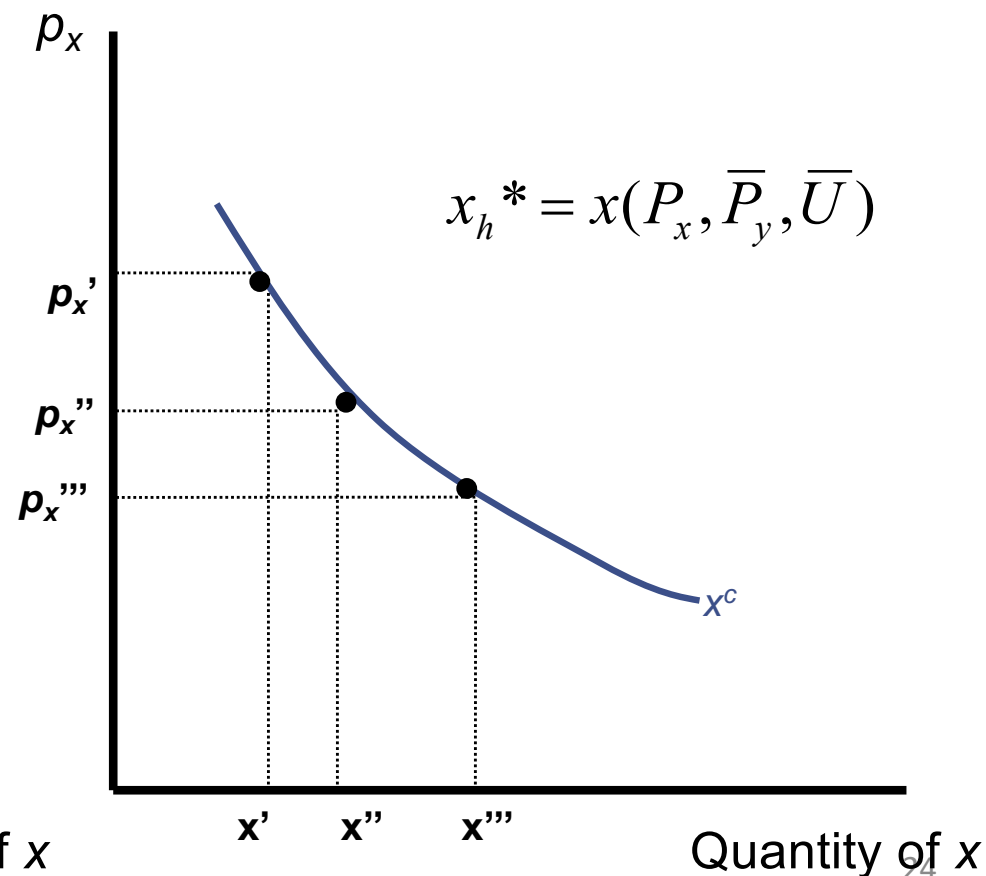
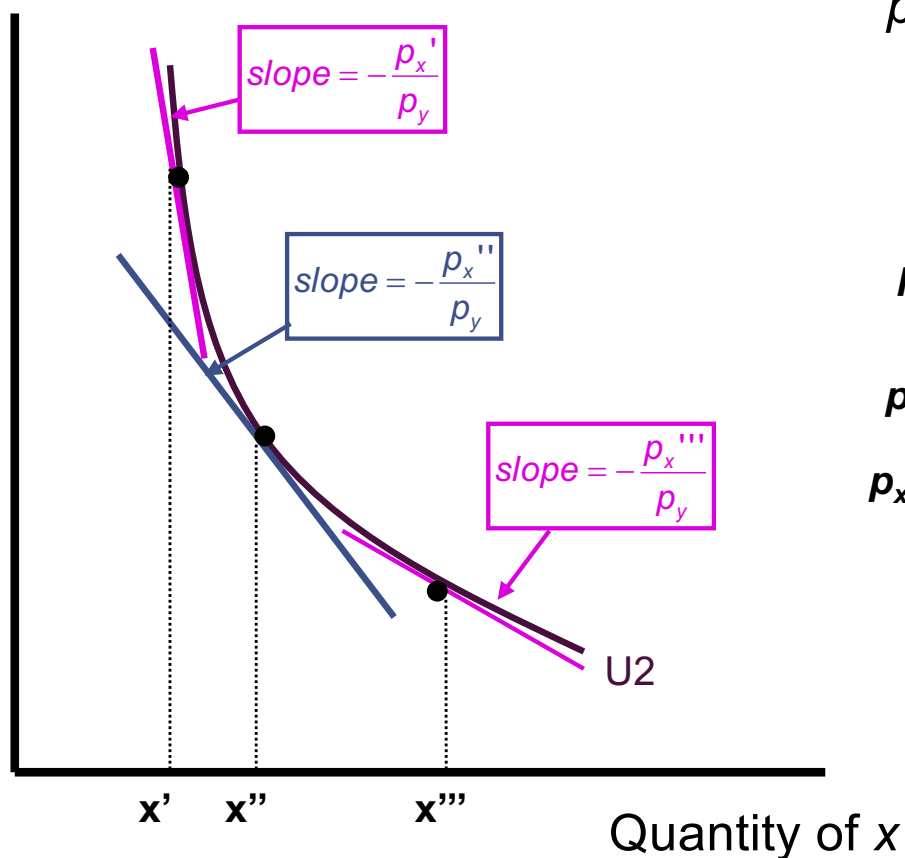


10. Individual Demand

Construction of a **compensated demand (Hicksian) curve**:

Shows the relationship between the price of a good and the quantity purchased on the assumption that **other prices and utility are held constant**

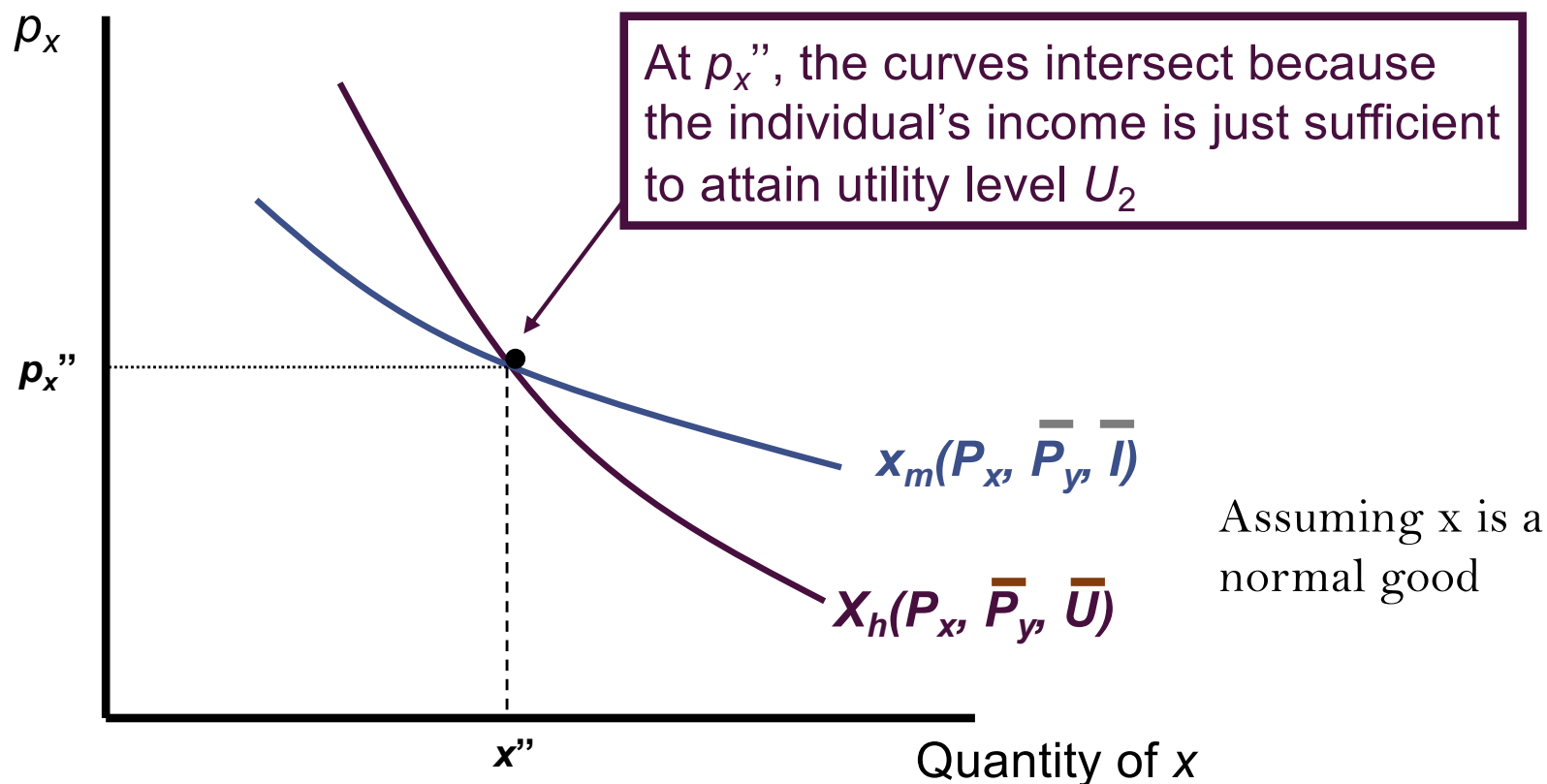
Quantity of y



10. Individual Demand

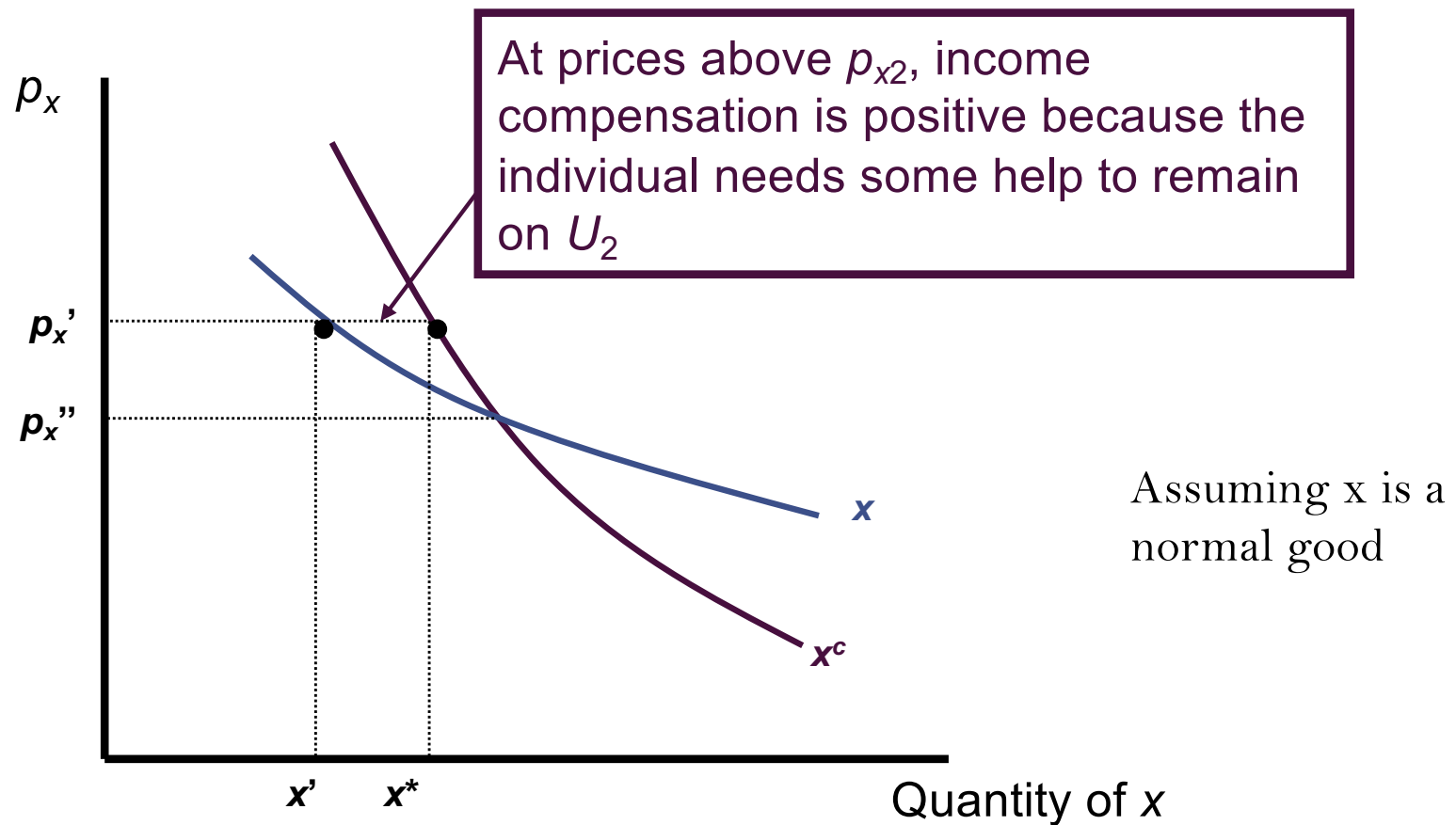
Comparison of Hicksian and Marshallian demand curves:

For a normal good, the compensated demand curve is less responsive to price changes than is the uncompensated demand curve



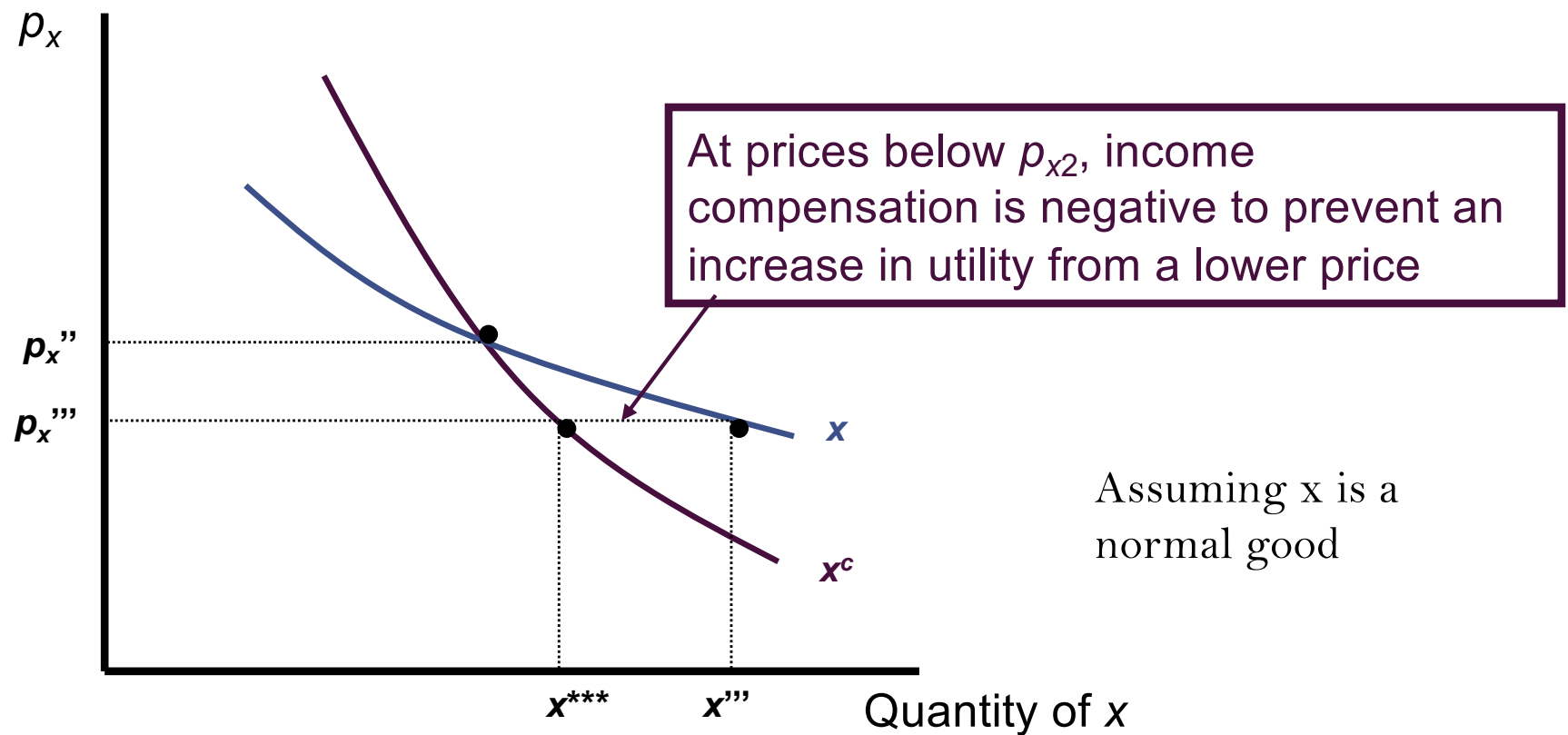
10. Individual Demand

Comparison of Hicksian and Marshallian demand curves:



10. Individual Demand

Comparison of Hicksian and Marshallian demand curves:



Demand Elasticities

Consumer Theory

11. Demand Elasticities

Most of the commonly used demand elasticities are derived from the Marshallian demand function $x(P_x, P_y, I)$

Price elasticity of demand:
$$\eta_{x_i, P_i} = \frac{\Delta x_i / x_i}{\Delta P_i / P_i} = \frac{\partial x_i}{\partial P_i} \frac{P_i}{x_i}$$

It should be negative for all goods (except for Giffen goods)

Elastic versus Inelastic demands

- If $\eta_{x_i, P_i} < -1 \rightarrow$ Elastic demand
- If $\eta_{x_i, P_i} = -1 \rightarrow$ Demand is unit elastic
- If $\eta_{x_i, P_i} > -1 \rightarrow$ Inelastic demand

**studies like crazy for his
microeconomic test**

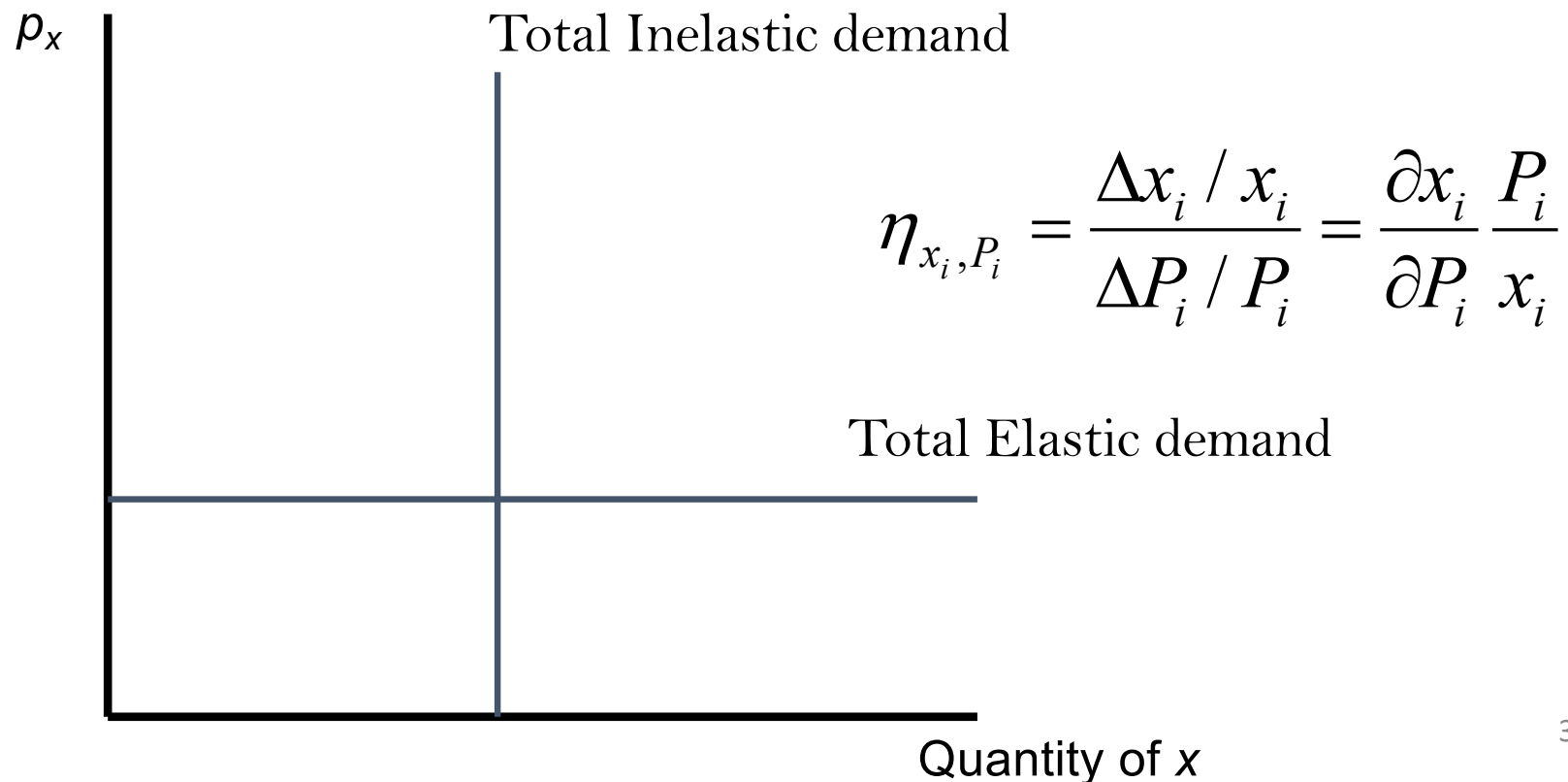
**has a perfectly inelastic study v/s
grade curve**

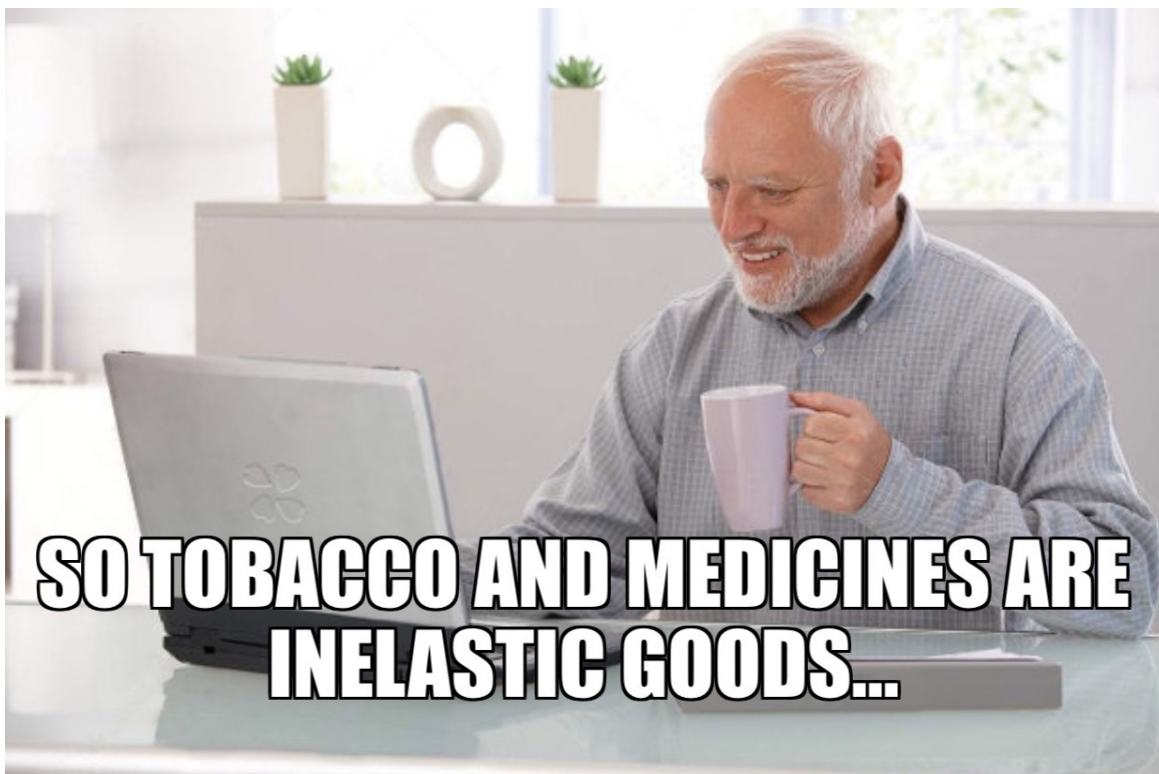
Author:
Nicolás
Errázuriz
(2021)

Memasik

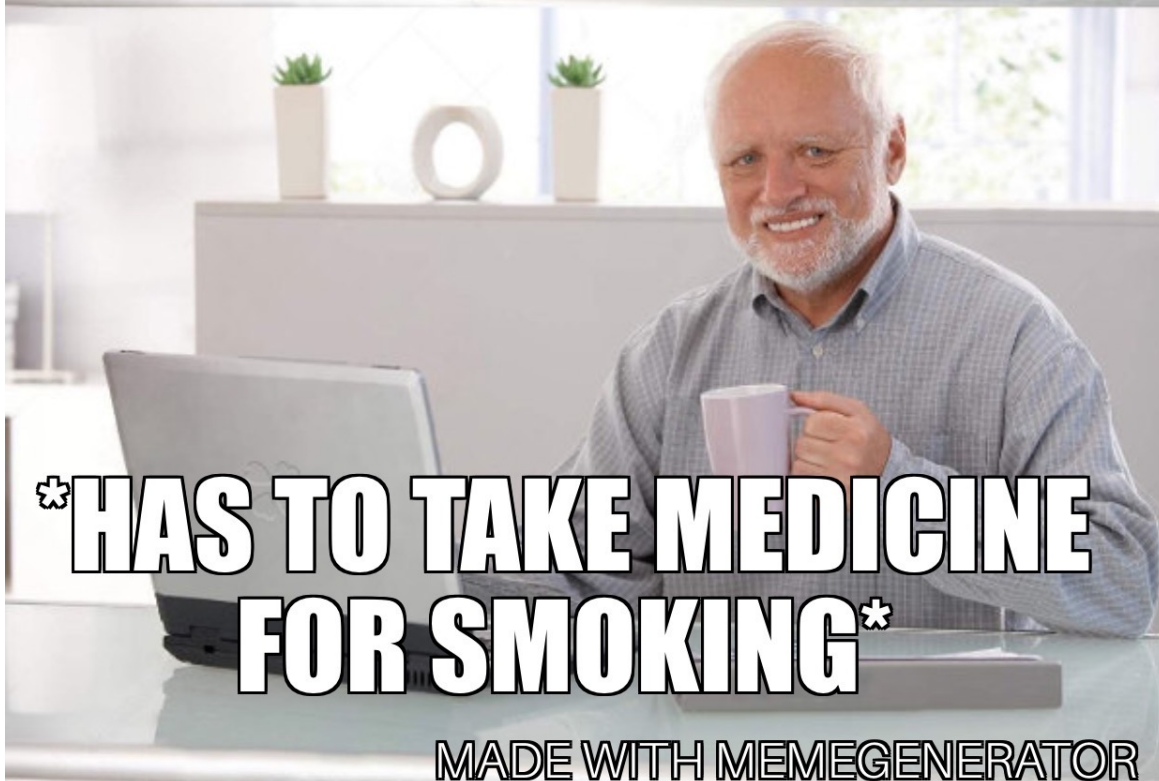
11. Demand Elasticities

- If $\eta_{x_i, P_i} < -1 \rightarrow$ Elastic demand
- If $\eta_{x_i, P_i} = -1 \rightarrow$ Demand is unit elastic
- If $\eta_{x_i, P_i} > -1 \rightarrow$ Inelastic demand





Author: Santiago Peirano (2023)



11. Demand Elasticities

Income elasticity of demand: $\eta_{x_i, I} = \frac{\Delta x_i / x_i}{\Delta I / I} = \frac{\partial x_i}{\partial I} \frac{I}{x_i}$

It should be positive for normal goods and negative for inferior goods

Cross-price elasticity of demand: $\eta_{x_i, P_j} = \frac{\Delta x_i / x_i}{\Delta P_j / P_j} = \frac{\partial x_i}{\partial P_j} \frac{P_j}{x_i}$

It should be positive for substitutes and negative for complements

Activity 1

In general, the demand curve should be more elastic: in the short run or in the long run?

- a. Short run
- b. Long run

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Wooclap code: **DKFTZP**



Slutsky equation (revisited)

We can rewrite the Slutsky equation in terms of the elasticity:

$$\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial P_i} = \frac{\partial h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))}{\partial P_i} - \frac{\partial x_i(P_1, \dots, P_n, I)}{\partial I} x_i(P_1, \dots, P_n, I)$$

If we multiply each side by P_i/x_i

$$\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial P_i} \frac{P_i}{x_i} = \frac{P_i}{x_i} \frac{\partial h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))}{\partial P_i} - \frac{\partial x_i(P_1, \dots, P_n, I)}{\partial I} \frac{I}{I} x_i(P_1, \dots, P_n, I) \frac{P_i}{x_i}$$

$$\eta_{x_i, P_i} = \eta_{x_i, P_i} \Big|_{U=\bar{U}} - \eta_{x_i, I} x_i(P_1, \dots, P_n, I) \frac{P_i}{I}$$

$$\eta_{x_i, P_i} = \eta_{x_i, P_i} \Big|_{U=\bar{U}} - \eta_{x_i, I} S_i$$

Market Demand

Consumer Theory

12. Market Demand

We want to **add (aggregate) the demand functions** from several people to reflect ALL individuals in a marketplace

In terms of notation, please note that subscript i will represent individuals from 1 to n .

Market demand for

$$X = \sum_{i=1}^n x_i(P_x, P_y, I_i)$$

12. Market Demand

Market demand for

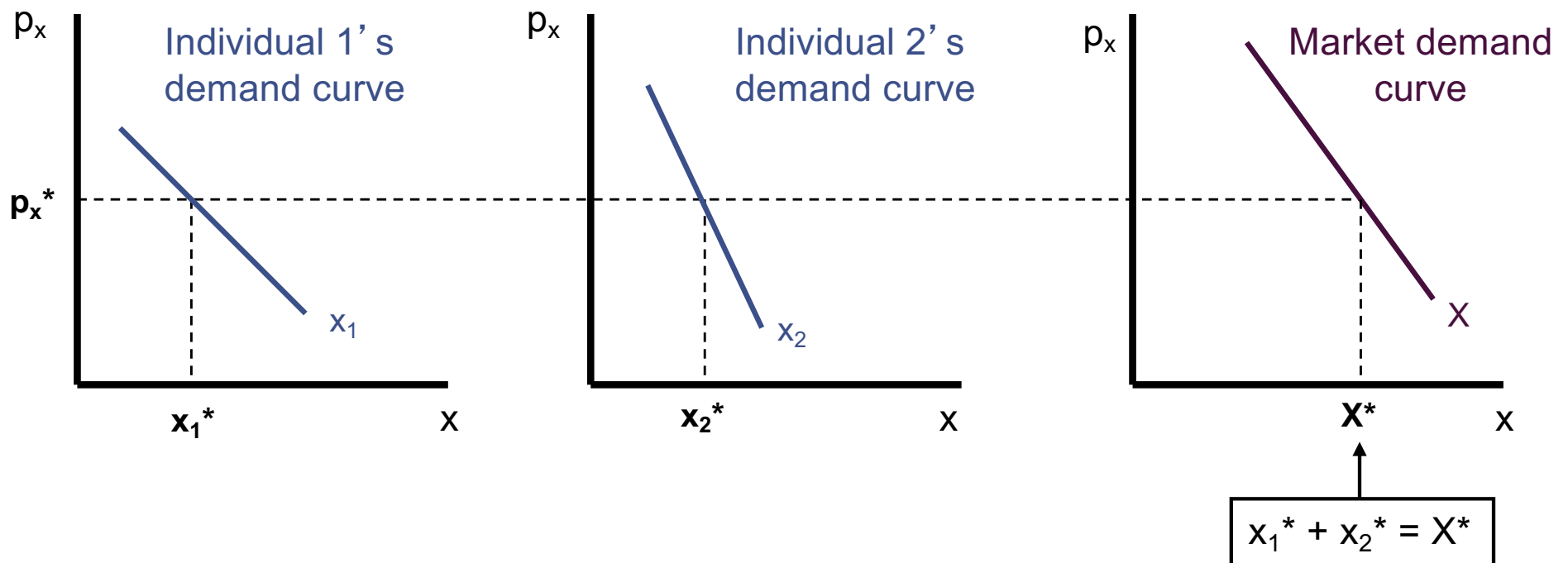
$$X = \sum_{i=1}^n x_i(P_x, P_y, I_i)$$

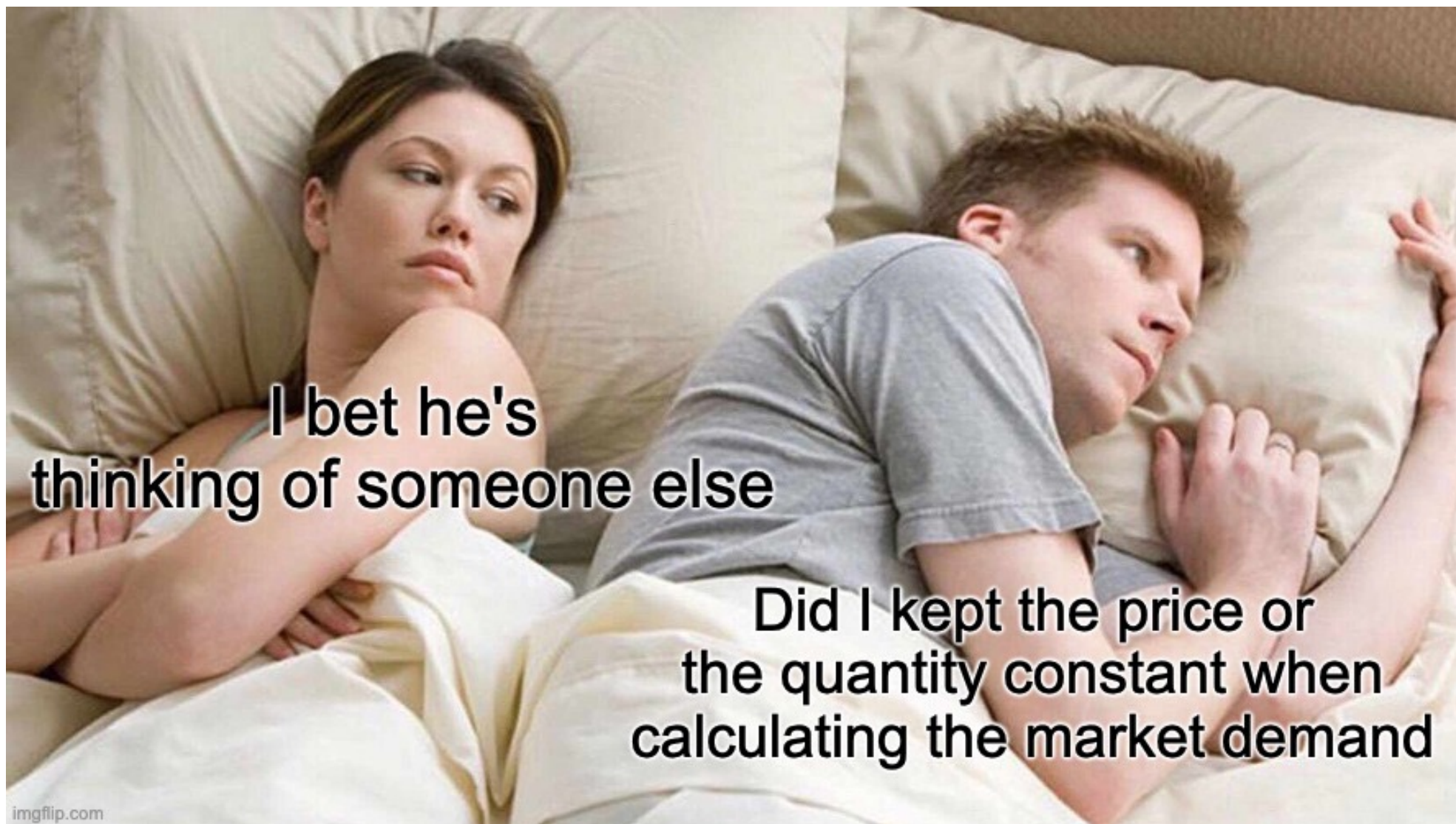
Assumptions:

- Everyone in this marketplace faces the same prices
- Each person's income enters into his or her own specific demand function → Market demand depends not only on the total income of all market participants, but also on how that income is distributed among consumers
- We are using uppercase X to refer to market demand

12. Market Demand

Construction of a market demand curve from individual demand curves





Author: Catalina Miranda (2023)

Activity 2

Shifts in the market demand curves:

Market demand for football fields might shift to the right (↗) because....

- a) Population decrease
- b) Decrease in the price of the football shoes
- c) Increase in the construction cost of the football fields
- d) Increase in the price of the football fields



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Wooclap code: **DKFTZP**

12. Market Demand

Market demand shifts to ↗ when:

- Income increases (normal goods)
- Prices of substitutes increase
- Price of complements decrease
- Population increases

$$X = \sum_{i=1}^n x_i(P_x, P_y, I_i)$$

Activity 3

In the chocolate market, we can distinguish two types of consumers: fans of chocolates versus normal people that like chocolates.

For the fans, their individual demand curve is given by

$$Ch_F = 30 - P_{Ch},$$

For the rest, their individual demand curve is given by:

$$Ch_N = 30 - 3P_{Ch},$$

where Ch is the quantity of chocolate consumed, P_{ch} is the price of chocolate. Assume that the market has 10 fan consumers of chocolate and 200 normal consumers of chocolate.

Determine the **total demand for chocolate**.

Hint: Graph the individual demand curves before attacking the problem mathematically.

Individual Activity

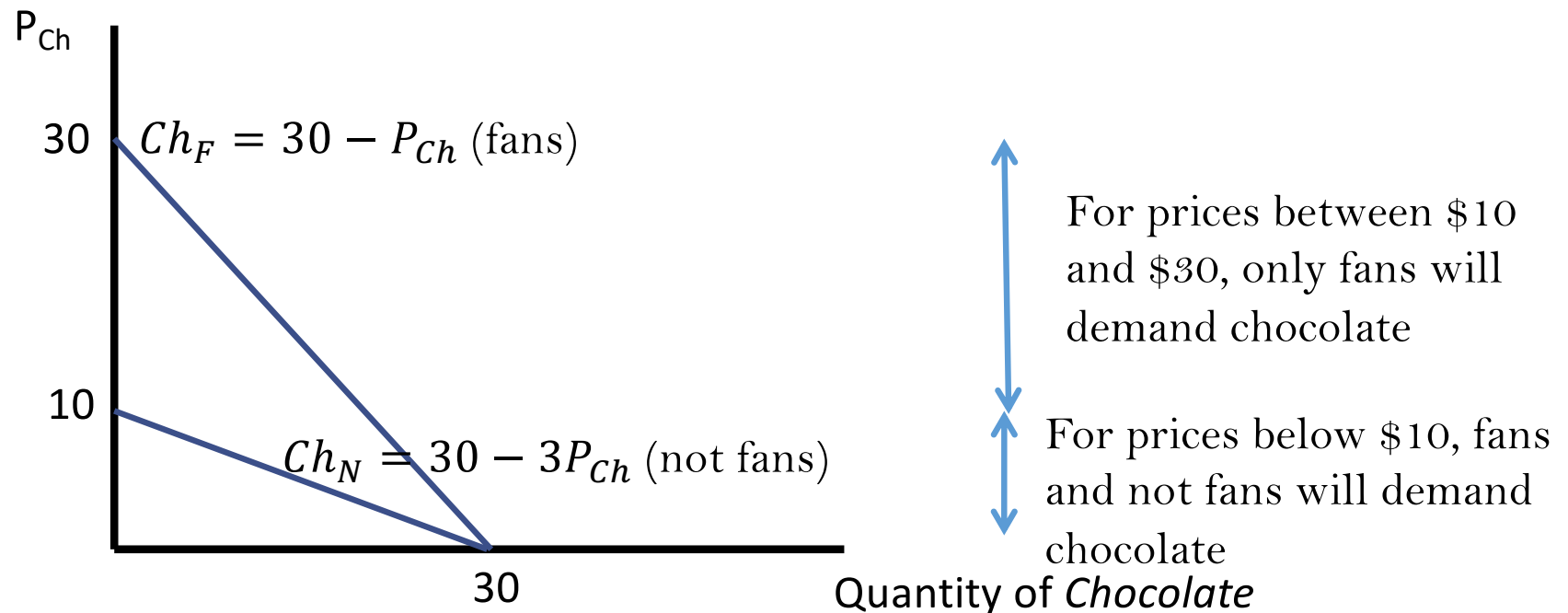
In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.

Activity 3: Solution

First, we should graph both demands in order to know how to sum them horizontally:



Activity 3: Solution

When price is between 10 and 30, only the 10 fans will demand chocolate:

$$Ch = 10(30 - P_{ch}) = 300 - 10P_{ch}, \text{ if } 10 < P_{ch} < 30$$

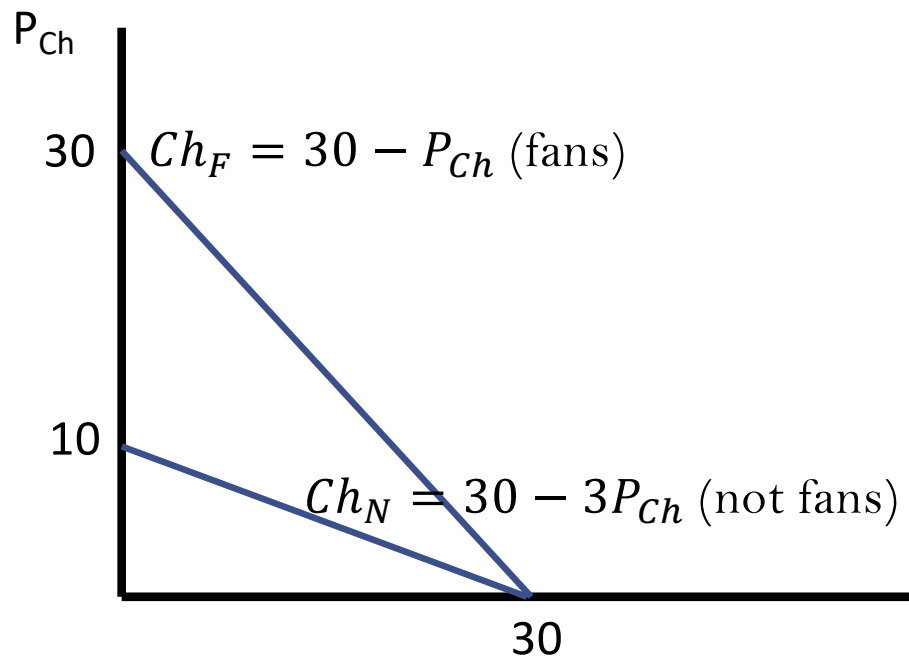
For prices between 0 and 10, fans and no fans will demand:

$$Ch = 10(30 - P_{ch}) + 200(30 - 3P_{ch})$$

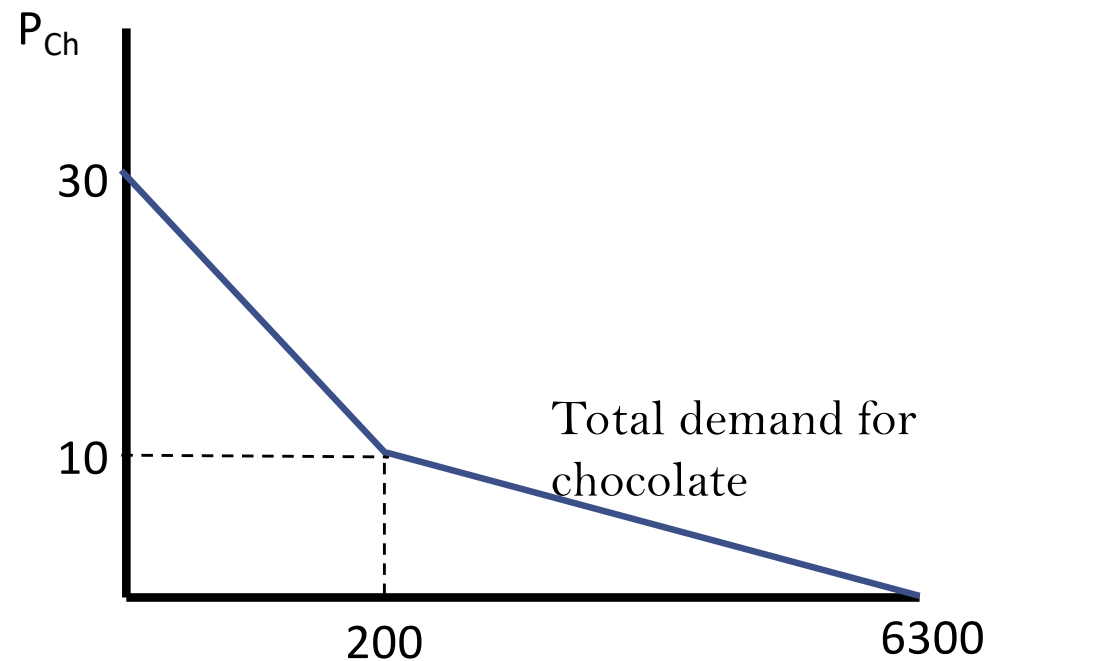
$$Ch = 6300 - 610P_{ch}, \text{ if } 0 < P_{ch} \leq 10$$

$$Ch = \begin{cases} 6300 - 610P_{ch} & \text{if } 0 < P_{ch} \leq 10 \\ 300 - 10P_{ch} & \text{if } 10 < P_{ch} \leq 30 \\ 0 & \text{otherwise} \end{cases}$$

Activity 3: Solution



Quantity of *Chocolate*



Quantity of *Chocolate*